Extra Course C: Antenna Basics and Feed Lines

Isotropic radiator, gain references (dBi vs dBd), ERP and EIRP math, ground systems, transmission-line behavior, velocity factor.

Groups E9A + E9F · 24 pool questions · 30-45 minute presentation · Foundation for Bites #16, #18 · Practice exams at eham.net/exams

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Part 1: Antenna Basics (E9A)

The Isotropic Radiator

An isotropic radiator is a hypothetical, lossless antenna having equal radiation intensity in all directions, used as a reference for antenna gain (E9A01).

You cannot build one. It is a mathematical reference: imagine a point source that radiates uniformly into the entire sphere around it. Real antennas concentrate power in some directions, so their gain in dBi (decibels over isotropic) tells you how much louder they are than that ideal sphere in their best direction.

There is a second reference too: **dBd**, decibels over a half-wave dipole. A half-wave dipole has **2.15 dB gain over isotropic** (because it pulls a little power out of the polar nulls and into broadside). The pool tests the conversion (E9A12):

Converting dBi → dBd: subtract 2.15. An antenna with 6 dBi gain has **6 − 2.15 ≈ 3.85 dBd** (E9A12).
Memory aid: dBi is always the bigger number for the same antenna.

Antenna Efficiency, Radiation Resistance, And Ground Losses

A real antenna's terminals see two kinds of "resistance" in series: **radiation resistance** (the power that actually leaves as RF) and **loss resistance** (power that becomes heat in conductors, ground, coil cores).

Antenna efficiency = radiation resistance / total resistance (E9A09).

Question	Answer	Q
Which factor affects an antenna's feed point impedance?	**Antenna height** (E9A04). Height above ground changes the ground-image coupling, which moves Z. The trap answers (transmission line length, tuner setting, input power) do **not** change the antenna's actual Z.	E9A04
What is "ground gain"?	**An increase in signal strength from ground reflections in the environment of the antenna** (E9A05). Reflected energy adds in-phase with the direct signal at certain angles, raising the apparent gain.	E9A05
Improve efficiency of a ground-mounted 1/4 λ vertical?	**Install a ground radial system** (E9A10). Radials reduce the loss-resistance current path through lossy soil.	E9A10
What determines ground losses for an HF ground-mounted vertical?	**Soil conductivity** (E9A11). Wet, mineral-rich soil is far better than dry sand.	E9A11
Which band has the **smallest** first Fresnel zone?	**5.8 GHz** (E9A08). Higher frequency = shorter wavelength = smaller Fresnel zone radius for the same path length.	E9A08

ERP And EIRP Math

Effective Radiated Power (ERP) and **Effective Isotropic Radiated Power (EIRP)** are how much apparent power the antenna throws in its main lobe, accounting for losses upstream and gain at the antenna.

General formula (in dB): $P_{ERP} = P_{TX}$ − (cable, duplexer, circulator losses) + (antenna gain).

ERP vs EIRP:

- **ERP** uses antenna gain in **dBd** (over a dipole).
- **EIRP** uses antenna gain in **dBi** (over isotropic).
- EIRP is always **2.15 dB higher** than ERP for the same antenna.

Term for "total radiated power that takes into account all gains and losses": Effective Radiated Power (E9A03).

Final conversion from dBm or dBW back to watts: power doubles every +3 dB, ×10 every +10 dB.

E9A02 worked: 150 W TX, 2 dB feed line loss, 2.2 dB duplexer loss, 7 dBd antenna gain.
Net dB change = −2 − 2.2 + 7 = **+2.8 dB**.
ERP = 150 W × $10^{0.28}$ ≈ 150 × 1.905 ≈ **286 W**.

E9A06 worked: 200 W TX, 4 dB feedline + 3.2 dB duplexer + 0.8 dB circulator + 10 dBd antenna.
Net = −4 − 3.2 − 0.8 + 10 = **+2.0 dB**. ERP = 200 × $10^{0.20}$ ≈ 200 × 1.585 ≈ **317 W**.

E9A07 worked (EIRP): 200 W TX, 2 dB feed + 2.8 dB duplexer + 1.2 dB circ + 7 dBi antenna.
Net = −2 − 2.8 − 1.2 + 7 = **+1.0 dB**. EIRP = 200 × $10^{0.10}$ ≈ 200 × 1.259 ≈ **252 W**.

Part 2: Feed Lines (E9F)

Velocity Factor And Electrical Length

Radio waves travel slower in a cable than in air because of the dielectric.

Velocity factor (VF) = velocity of a wave in the line / velocity of light in a vacuum (E9F01). Always less than 1. Common values: foam coax 0.85, solid PE coax 0.66, air-dielectric ladder line 0.95+.

The biggest effect on VF is the insulating dielectric material (E9F02). Different plastics slow the wave by different amounts.

Why is the electrical length of a coax longer than its physical length? Because **electromagnetic waves move more slowly in coax than in air** (E9F03). Said another way: if you need an electrical quarter wavelength, the physical coax you cut is shorter than a free-space quarter wave by the velocity factor. For a fixed physical length, that piece is still electrically longer than its physical length (consistent with E9F03).

Length Calculation

E9F06 worked: physical length of an **air-insulated parallel conductor** line that is electrically 1/2 λ at 14.10 MHz.
λ = 300 / f(MHz) ≈ 300 / 14.10 ≈ 21.28 m.
Half wavelength = 21.28 / 2 ≈ 10.64 m.
Air dielectric → VF ≈ 1.0, so physical length ≈ **10.6 m** (E9F06).

Open Versus Shorted Stubs

A length of transmission line shorted (or opened) at the far end transforms what the generator sees in predictable ways. These are exam-test favorites:

Stub length	Far end	What the generator sees	Q
1/4 λ	Shorted	**Very high impedance** (open-circuit-like). (E9F09)	E9F09
1/4 λ	Open	**Very low impedance** (short-circuit-like). (E9F12)	E9F12
1/2 λ	Shorted	**Very low impedance** (still shorted-looking). (E9F04)	E9F04
1/8 λ	Shorted	**Inductive reactance** (E9F10).	E9F10
1/8 λ	Open	**Capacitive reactance** (E9F11).	E9F11

Memory pattern: A quarter-wave line **inverts** what is on the far end (short → open, open → short). A half-wave line **preserves** what is on the far end. Eighth-wave lines look reactive (lossless reactance) – shorted gives inductive, open gives capacitive.

Coax Vs Parallel Conductor And Microstrip

Question	Answer	Q
Parallel-conductor line vs plastic-dielectric coax	**Lower loss** (E9F07). Open-wire has air dielectric and less conductor surface in the field, so its loss is far below plastic-coax loss.	E9F07
Foam-dielectric coax vs solid coax (same all-other-params)	Lower max operating voltage + lower loss per length + higher velocity factor → **all of the above** (E9F08). Foam is mostly air, so all three follow.	E9F08
What is microstrip?	**Precision printed-circuit conductors above a ground plane that provide constant-impedance interconnects at microwave frequencies** (E9F05).	E9F05

Mental Cheat Sheet

- Isotropic radiator = ideal point source, used as gain reference.
- dBi vs dBd: subtract 2.15 to get dBd from dBi.
- Antenna efficiency = radiation R / total R.
- Antenna height changes feed-point impedance.
- Ground gain = signal boost from ground reflections.
- Vertical antenna efficiency improved by ground radials.
- HF vertical ground loss is dominated by soil conductivity.
- 5.8 GHz has the smallest Fresnel zone.
- ERP / EIRP = TX power, minus all losses, plus antenna gain (in dBd / dBi).
- Velocity factor = signal speed in line / c. Set by the dielectric.
- Electrical length of coax is longer than physical length because the wave is slow.
- Quarter-wave stub **inverts** far-end; half-wave **preserves** it.
- 1/8 λ shorted = inductive; 1/8 λ open = capacitive.
- Parallel-conductor line has lower loss than plastic coax.
- Foam coax has lower V max + lower loss + higher VF than solid coax.
- Microstrip = PCB trace over a ground plane → constant impedance at μwave.

Common Traps

- Isotropic radiator is **hypothetical**, not a real calibrated antenna.

- Feed-point Z is changed by **height**, not by feed-line length, tuner setting, or power level.
- ERP uses dBd; EIRP uses dBi. Read the question's units carefully.
- Velocity factor is set by the **dielectric**, not the impedance or length.
- A 1/2 λ shorted stub looks like a **short** to the generator (not an open).
- A 1/4 λ open stub looks like a **short** to the generator (not the line's Z_0).
- 1/8 λ shorted → **inductive**; 1/8 λ open → **capacitive**. Mix-up is the test's favorite trap here.
- "Why is electrical length longer than physical length?" → because the wave is **slower** in coax than in air.

Takeaway: Gain references (dBi vs dBd) and loss/gain math get you ERP and EIRP. A vertical's efficiency depends on radials and soil. On the feed-line side, velocity factor governs electrical length. Open and shorted stubs are predictable: quarter wave inverts the far end, half wave preserves it, eighth wave looks reactive. Open-wire beats plastic coax on loss; foam coax beats solid on loss and VF.

Course C Question Bank (E9A + E9F)

E9A01. What is an isotropic radiator?
A. A calibrated, unidirectional antenna used to make precise antenna gain measurements.
B. An omnidirectional, horizontally polarized, precisely calibrated antenna used to make field measurements of antenna gain.
C. A hypothetical, lossless antenna having equal radiation intensity in all directions used as a reference for antenna gain.
D. A spacecraft antenna used to direct signals toward Earth.

E9A02. What is the effective radiated power (ERP) of a repeater station with 150 watts transmitter power output, 2 dB feed line loss, 2.2 dB duplexer loss, and 7 dBd antenna gain?
A. 469 watts.
B. 78.7 watts.
C. 420 watts.
D. 286 watts.

E9A03. What term describing total radiated power takes into account all gains and losses?
A. Power factor.
B. Half-power bandwidth.
C. Effective radiated power.
D. Apparent power.

E9A04. Which of the following factors affect the feed point impedance of an antenna?
A. Transmission line length.
B. Antenna height.
C. The settings of an antenna tuner at the transmitter.
D. The input power level.

E9A05. What does the term "ground gain" mean?
A. The change in signal strength caused by grounding the antenna.
B. The gain of the antenna with respect to a dipole at ground level.
C. To force net gain to 0 dB by grounding part of the antenna.
D. An increase in signal strength from ground reflections in the environment of the antenna.

E9A06. What is the effective radiated power (ERP) of a repeater station with 200 watts transmitter power output, 4 dB feed line loss, 3.2 dB duplexer loss, 0.8 dB circulator loss, and 10 dBd antenna gain?
A. 317 watts.
B. 2,000 watts.
C. 126 watts.
D. 300 watts.

E9A07. What is the effective isotropic radiated power (EIRP) of a repeater station with 200 watts transmitter power output, 2 dB feed line loss, 2.8 dB duplexer loss, 1.2 dB circulator loss, and 7 dBi antenna gain?
A. 159 watts.
B. 252 watts.
C. 632 watts.
D. 63.2 watts.

E9A08. Which frequency band has the smallest first Fresnel zone?
A. 5.8 GHz.
B. 3.4 GHz.
C. 2.4 GHz.
D. 900 MHz.

E9A09. What is antenna efficiency?
A. Radiation resistance divided by transmission resistance.
B. Radiation resistance divided by total resistance.
C. Total resistance divided by radiation resistance.
D. Effective radiated power divided by transmitter output.

E9A10. Which of the following improves the efficiency of a ground-mounted quarter-wave vertical antenna?
A. Installing a ground radial system.
B. Isolating the coax shield from ground.
C. Shortening the radiating element.
D. All these choices are correct.

E9A11. Which of the following determines ground losses for a ground-mounted vertical antenna operating on HF?
A. The standing wave ratio.
B. Distance from the transmitter.
C. Soil conductivity.
D. Take-off angle.

E9A12. How much gain does an antenna have compared to a half-wavelength dipole if it has 6 dB gain over an isotropic radiator?
A. 3.85 dB.
B. 6.0 dB.
C. 8.15 dB.
D. 2.79 dB.

E9F01. What is the velocity factor of a transmission line?
A. The ratio of its characteristic impedance to its termination impedance.
B. The ratio of its termination impedance to its characteristic impedance.
C. The velocity of a wave in the transmission line multiplied by the velocity of light in a vacuum.
D. The velocity of a wave in the transmission line divided by the velocity of light in a vacuum.

E9F02. Which of the following has the biggest effect on the velocity factor of a transmission line?
A. The characteristic impedance.
B. The transmission line length.
C. The insulating dielectric material.
D. The center conductor resistivity.

E9F03. Why is the electrical length of a coaxial cable longer than its physical length?
A. Skin effect is less pronounced in the coaxial cable.
B. Skin effect is more pronounced in the coaxial cable.
C. Electromagnetic waves move faster in coaxial cable than in air.
D. Electromagnetic waves move more slowly in a coaxial cable than in air.

E9F04. What impedance does a 1/2-wavelength transmission line present to an RF generator when the line is shorted at the far end?
A. Very high impedance.
B. Very low impedance.
C. The same as the characteristic impedance of the line.
D. The same as the output impedance of the RF generator.

E9F05. What is microstrip?
A. Special shielding material designed for microwave frequencies.
B. Miniature coax used for low power applications.
C. Short lengths of coax mounted on printed circuit boards to minimize time delay between microwave circuits.
D. Precision printed circuit conductors above a ground plane that provide constant impedance interconnects at microwave frequencies.

E9F06. What is the approximate physical length of an air-insulated, parallel conductor transmission line that is electrically 1/2 wavelength long at 14.10 MHz?
A. 7.0 meters.
B. 8.5 meters.
C. 10.6 meters.
D. 13.3 meters.

E9F07. How does parallel conductor transmission line compare to coaxial cable with a plastic dielectric?
A. Lower loss.
B. Higher SWR.
C. Smaller reflection coefficient.
D. Lower velocity factor.

E9F08. Which of the following is a significant difference between foam dielectric coaxial cable and solid dielectric coaxial cable, assuming all other parameters are the same?
A. Foam dielectric coaxial cable has lower safe maximum operating voltage.
B. Foam dielectric coaxial cable has lower loss per unit of length.
C. Foam dielectric coaxial cable has higher velocity factor.
D. All these choices are correct.

E9F09. What impedance does a 1/4-wavelength transmission line present to an RF generator when the line is shorted at the far end?
A. Very high impedance.
B. Very low impedance.
C. The same as the characteristic impedance of the transmission line.
D. The same as the generator output impedance.

E9F10. What impedance does a 1/8-wavelength transmission line present to an RF generator when the line is shorted at the far end?
A. A capacitive reactance.
B. The same as the characteristic impedance of the line.
C. An inductive reactance.
D. Zero.

E9F11. What impedance does a 1/8-wavelength transmission line present to an RF generator when the line is open at the far end?
A. The same as the characteristic impedance of the line.
B. An inductive reactance.
C. A capacitive reactance.
D. Infinite.

E9F12. What impedance does a 1/4-wavelength transmission line present to an RF generator when the line is open at the far end?
A. The same as the characteristic impedance of the line.
B. The same as the input impedance to the generator.
C. Very high impedance.
D. Very low impedance.

Answer Key

E9A01: C	E9A02: D	E9A03: C	E9A04: B	E9A05: D
E9A06: A	E9A07: B	E9A08: A	E9A09: B	E9A10: A
E9A11: C	E9A12: A	E9F01: D	E9F02: C	E9F03: D
E9F04: B	E9F05: D	E9F06: C	E9F07: A	E9F08: D
E9F09: A	E9F10: C	E9F11: C	E9F12: D	